

COMP5425 26s1 Final Practice

USYD 各科往届题库



W1 Introduction and Retrieval System Design

Pitfall: metadata-only design; Focus: full pipeline; **Pitfall: no evaluation;** Focus: latency + relevance

Q1. MCQ

Which statement is most appropriate for multimedia retrieval pipeline?

- (A) A two-stage pipeline with cheap candidate generation and expensive reranking.
- (B) A single exact scan over all raw videos for every query.
- (C) Only manual tags, because content features are never useful.
- (D) A database sorted alphabetically by file name.

Q2. Short Answer

Explain multimedia retrieval pipeline using the course pipeline language. Your answer must mention feature extraction, indexing, similarity, feedback, evaluation.

Q3. Long Question Semester 1 2021 Final (modified)

Adapt the source-style question to multimedia retrieval pipeline. Answer all parts:

1. State the underlying principle.
2. Describe the implementation steps.
3. Give two advantages and two limitations.
4. Propose one concrete improvement for a modern multimedia retrieval setting.

Q4. True/False

True or false: In Introduction and Retrieval System Design, higher numerical score always means a better retrieval result. Justify your answer with one counterexample.

Q5. Short Answer

A student makes this mistake: confusing metadata search with content-based search. Explain why it is wrong and give a corrected approach.

Q6. Table/Diagram

Construct a two-column comparison table for multimedia retrieval pipeline: left column = method/component, right column = retrieval benefit and limitation.

Q7. MCQ

A retrieval system for multimedia retrieval pipeline performs poorly on a new domain. What is the best first diagnostic step?

- (A) Delete the evaluation set and report only training results.
- (B) Add more algorithms without checking data or metrics.
- (C) Inspect false positives and false negatives by feature, query type and user intent.
- (D) Assume the metric is wrong because the model is modern.

Q8. Short Answer

Compare an exact method and an approximate or heuristic method relevant to multimedia retrieval pipeline. When would the approximate method be justified?

Q9. Calculation

A system returns 8 relevant items in the top 10; the collection has 20 relevant items. Compute precision@10 and recall@10.

Q10. Long Question

Design retrieval for a university multimedia library containing PDFs, lecture videos and slide images.

1. Define two user groups and their query types.
2. Choose text, visual and metadata representations.
3. Choose a two-stage indexing/ranking strategy.
4. Describe one feedback or personalisation mechanism.
5. Specify evaluation metrics and one operational limitation.



W2 Text Information Retrieval

Pitfall: raw TF only; Focus: postings + positions; **Pitfall: removing domain terms;** Focus: cosine/IDF

Q1. MCQ

Which statement is most appropriate for document preprocessing and inverted index?

- (A) A dense term-document matrix stored without compression.
- (B) An inverted index with postings containing document ids and positions.
- (C) A random sample of documents with no dictionary.
- (D) A PageRank vector over terms.

Q2. Short Answer

Explain document preprocessing and inverted index using the course pipeline language. Your answer must mention tokenisation, stemming, stop words, postings, TF-IDF.

Q3. Long Question Semester 1 2024 Final (modified)

Adapt the source-style question to document preprocessing and inverted index. Answer all parts:

1. State the underlying principle.
2. Describe the implementation steps.
3. Give two advantages and two limitations.
4. Propose one concrete improvement for a modern multimedia retrieval setting.

Q4. True/False

True or false: In Text Information Retrieval, higher numerical score always means a better retrieval result. Justify your answer with one counterexample.

Q5. Short Answer

A student makes this mistake: using raw term frequency without IDF or normalisation. Explain why it is wrong and give a corrected approach.

Q6. Table/Diagram

Construct a two-column comparison table for document preprocessing and inverted index: left column = method/component, right column = retrieval benefit and limitation.

Q7. MCQ

A retrieval system for document preprocessing and inverted index performs poorly on a new domain. What is the best first diagnostic step?

- (A) Delete the evaluation set and report only training results.
- (B) Add more algorithms without checking data or metrics.
- (C) Assume the metric is wrong because the model is modern.
- (D) Inspect false positives and false negatives by feature, query type and user intent.

Q8. Short Answer

Compare an exact method and an approximate or heuristic method relevant to document preprocessing and inverted index. When would the approximate method be justified?

Q9. Calculation

Three documents have term counts for term 'image' of 2, 0, and 6. If $N = 10$ and $df = 2$, compute unsmoothed $idf = \log_{10}(N/df)$ and TF-IDF for the two nonzero documents using raw TF.

Q10. Long Question

Design the text retrieval core for 200,000 course documents where students search for exact phrases and broad topics.

1. Define preprocessing choices and what not to remove.
2. Specify postings fields needed for phrase/proximity search.
3. Choose TF-IDF/BM25-style ranking signals.
4. Add query expansion or relevance feedback safely.
5. Evaluate ranking quality and diagnose one failure mode.



W3 Web Information Retrieval

Pitfall: ignore outdegree; **Focus:** teleportation; **Pitfall:** HITS topic drift; **Focus:** anchor text

Q1. MCQ

Which statement is most appropriate for PageRank and HITS?

- (A) Give every incoming link the full score of the source page.
- (B) Ignore teleportation and dangling pages in all cases.
- (C) Divide each page score by its number of outgoing links before distributing it.
- (D) Normalise by page title length.

Q2. Short Answer

Explain PageRank and HITS using the course pipeline language. Your answer must mention crawler, link graph, PageRank, HITS, anchor text.

Q3. Long Question Semester 1 2023 Final (modified)

Adapt the source-style question to PageRank and HITS. Answer all parts:

1. State the underlying principle.
2. Describe the implementation steps.
3. Give two advantages and two limitations.
4. Propose one concrete improvement for a modern multimedia retrieval setting.

Q4. True/False

True or false: In Web Information Retrieval, higher numerical score always means a better retrieval result. Justify your answer with one counterexample.

Q5. Short Answer

A student makes this mistake: forgetting outdegree and teleportation in PageRank. Explain why it is wrong and give a corrected approach.

Q6. Table/Diagram

Construct a two-column comparison table for PageRank and HITS: left column = method/component, right column = retrieval benefit and limitation.

Q7. MCQ

A retrieval system for PageRank and HITS performs poorly on a new domain. What is the best first diagnostic step?

- (A) Inspect false positives and false negatives by feature, query type and user intent.
- (B) Delete the evaluation set and report only training results.
- (C) Add more algorithms without checking data or metrics.
- (D) Assume the metric is wrong because the model is modern.

Q8. Short Answer

Compare an exact method and an approximate or heuristic method relevant to PageRank and HITS. When would the approximate method be justified?

Q9. Calculation

Pages A,B,C have links $A \rightarrow B,C$; $B \rightarrow C$; $C \rightarrow A$. With $d = 0.85$ and initial scores $1/3$, compute one PageRank update for A.

Q10. Long Question

Design web ranking for a small academic search engine that crawls lab pages, papers and project sites.

1. Define crawling and canonicalisation requirements.
2. Specify how PageRank and anchor text enter ranking.
3. Explain when HITS would be computed for a query.
4. Mitigate topic drift, spam or link farms.
5. Evaluate relevance, freshness and link-quality impact.



W4 Multimedia Basics

Pitfall: no retrieval impact; Focus: sampling vs quantisation; **Pitfall: aliasing;** Focus: acquisition quality

Q1. MCQ

Which statement is most appropriate for digitisation and media representation?

- (A) Quantisation increases the number of possible amplitude values.
- (B) Higher compression always improves retrieval features.
- (C) Frame rate has no effect on video retrieval.
- (D) Sampling too slowly relative to signal frequency can create aliasing.

Q2. Short Answer

Explain digitisation and media representation using the course pipeline language. Your answer must mention sampling, quantisation, aliasing, histogram equalisation.

Q3. Long Question Semester 1 2021 Final (modified)

Adapt the source-style question to digitisation and media representation. Answer all parts:

1. State the underlying principle.
2. Describe the implementation steps.
3. Give two advantages and two limitations.
4. Propose one concrete improvement for a modern multimedia retrieval setting.

Q4. True/False

True or false: In Multimedia Basics, higher numerical score always means a better retrieval result. Justify your answer with one counterexample.

Q5. Short Answer

A student makes this mistake: describing signal processing without retrieval consequences. Explain why it is wrong and give a corrected approach.

Q6. Table/Diagram

Construct a two-column comparison table for digitisation and media representation: left column = method/component, right column = retrieval benefit and limitation.

Q7. MCQ

A retrieval system for digitisation and media representation performs poorly on a new domain. What is the best first diagnostic step?

- (A) Delete the evaluation set and report only training results.
- (B) Inspect false positives and false negatives by feature, query type and user intent.
- (C) Add more algorithms without checking data or metrics.
- (D) Assume the metric is wrong because the model is modern.

Q8. Short Answer

Compare an exact method and an approximate or heuristic method relevant to digitisation and media representation. When would the approximate method be justified?

Q9. Calculation

An 8-level image has probabilities (0.1, 0.2, 0.3, 0.4, 0, 0, 0, 0) for levels 0–7. Compute the equalisation mapping for levels 0–3 using $s_k = \text{round}(7 \cdot CDF)$.

Q10. Long Question

Design media ingestion for a retrieval archive receiving images, audio and video from mixed devices.

1. Identify sampling/quantisation parameters to store.
2. Explain how compression artifacts affect features.
3. Choose normalisation/restoration steps before indexing.
4. State how metadata records acquisition quality.
5. Evaluate whether preprocessing improves retrieval.



W5 Image Retrieval Features

Pitfall: semantic gap; Focus: feature normalisation; **Pitfall: colour-only answer**; Focus: metric choice

Q1. MCQ

Which statement is most appropriate for content-based image retrieval?

- (A) CCV adds coarse spatial coherence information to colour counts.
- (B) Colour histogram fully captures object shape.
- (C) Texture descriptors require no normalisation.
- (D) Euclidean distance is always semantic similarity.

Q2. Short Answer

Explain content-based image retrieval using the course pipeline language. Your answer must mention colour histogram, CCV, texture, shape, feature normalisation.

Q3. Long Question Semester 1 2023 Final (modified)

Adapt the source-style question to content-based image retrieval. Answer all parts:

1. State the underlying principle.
2. Describe the implementation steps.
3. Give two advantages and two limitations.
4. Propose one concrete improvement for a modern multimedia retrieval setting.

Q4. True/False

True or false: In Image Retrieval Features, higher numerical score always means a better retrieval result. Justify your answer with one counterexample.

Q5. Short Answer

A student makes this mistake: treating visually similar low-level features as semantic equality. Explain why it is wrong and give a corrected approach.

Q6. Table/Diagram

The following pipeline is a compact CBIR design. Explain what each stage contributes and identify one likely failure point.



Q7. MCQ

A retrieval system for content-based image retrieval performs poorly on a new domain. What is the best first diagnostic step?

- (A) Delete the evaluation set and report only training results.
- (B) Add more algorithms without checking data or metrics.
- (C) Inspect false positives and false negatives by feature, query type and user intent.
- (D) Assume the metric is wrong because the model is modern.

Q8. Short Answer

Compare an exact method and an approximate or heuristic method relevant to content-based image retrieval. When would the approximate method be justified?

Q9. Calculation

For query vector $(0.5, 0.7, 0.2)$ and image vector $(0.3, 0.8, 0.4)$, compute Euclidean distance.

Q10. Long Question

Design CBIR for a museum collection where users upload an artwork photo to find visually related works.

1. Select colour, texture and shape/local features.
2. Explain feature normalisation and distance choice.
3. Choose an index and reranking step.
4. Add relevance feedback to reduce semantic gap.
5. Evaluate precision@k and failure cases such as lighting.



W6 CBIR Indexing and Feedback

Pitfall: unnormalised fusion; Focus: recall@k + latency; **Pitfall: exact search at scale;** Focus: ANN vs

Q1. MCQ

Which statement is most appropriate for similarity search and indexing?

- (A) LSH guarantees exact nearest neighbours with no false negatives.
- (B) LSH returns approximate candidates quickly by hashing similar vectors into shared buckets.
- (C) k-d trees improve indefinitely as dimension grows.
- (D) Distance matrices use sublinear storage.

Q2. Short Answer

Explain similarity search and indexing using the course pipeline language. Your answer must mention distance functions, k-d tree, LSH, relevance feedback, feature fusion.

Q3. Long Question Semester 1 2024 Final (modified)

Adapt the source-style question to similarity search and indexing. Answer all parts:

1. State the underlying principle.
2. Describe the implementation steps.
3. Give two advantages and two limitations.
4. Propose one concrete improvement for a modern multimedia retrieval setting.

Q4. True/False

True or false: In CBIR Indexing and Feedback, higher numerical score always means a better retrieval result. Justify your answer with one counterexample.

Q5. Short Answer

A student makes this mistake: mixing incomparable feature scales. Explain why it is wrong and give a corrected approach.

Q6. Table/Diagram

Construct a two-column comparison table for similarity search and indexing: left column = method/component, right column = retrieval benefit and limitation.

Q7. MCQ

A retrieval system for similarity search and indexing performs poorly on a new domain. What is the best first diagnostic step?

- (A) Delete the evaluation set and report only training results.
- (B) Add more algorithms without checking data or metrics.
- (C) Assume the metric is wrong because the model is modern.
- (D) Inspect false positives and false negatives by feature, query type and user intent.

Q8. Short Answer

Compare an exact method and an approximate or heuristic method relevant to similarity search and indexing. When would the approximate method be justified?

Q9. Calculation

Item X has colour distance 0.8 and texture distance 20. Item Y has colour distance 0.4 and texture distance 70. Given observed ranges colour 0–1 and texture 0–100, compute equal-weight fused distances and rank X/Y, where smaller fused distance is better.

Q10. Long Question

Design scalable similarity search for 30 million image embeddings plus handcrafted colour/texture features.

1. Choose exact or ANN candidate generation.
2. Explain feature-score normalisation and fusion.
3. Choose k-d tree, LSH or graph/vector index and justify.
4. Add feedback or reranking.
5. Evaluate recall@k, latency and memory.



W7 Video Retrieval

Pitfall: keyframe-only video; Focus: shot hierarchy; **Pitfall: motion ignored;** Focus: event-level recall

Q1. MCQ

Which statement is most appropriate for video hierarchy and shot representation?

- (A) A shot is always the same as a full movie.
- (B) Histogram change cannot help shot boundary detection.
- (C) A keyframe is a compact representative image for a shot.
- (D) Video summaries must include every frame.

Q2. Short Answer

Explain video hierarchy and shot representation using the course pipeline language. Your answer must mention shot boundary, keyframes, motion, scenes, summaries.

Q3. Long Question Semester 1 2024 Final (modified)

Adapt the source-style question to video hierarchy and shot representation. Answer all parts:

1. State the underlying principle.
2. Describe the implementation steps.
3. Give two advantages and two limitations.
4. Propose one concrete improvement for a modern multimedia retrieval setting.

Q4. True/False

True or false: In Video Retrieval, higher numerical score always means a better retrieval result. Justify your answer with one counterexample.

Q5. Short Answer

A student makes this mistake: using keyframes when temporal motion is essential. Explain why it is wrong and give a corrected approach.

Q6. Table/Diagram

Draw or complete a three-level video hierarchy for retrieval and state what should be indexed at each level: frame, shot, and scene/event.

Q7. MCQ

A retrieval system for video hierarchy and shot representation performs poorly on a new domain. What is the best first diagnostic step?

- (A) Inspect false positives and false negatives by feature, query type and user intent.
- (B) Delete the evaluation set and report only training results.
- (C) Add more algorithms without checking data or metrics.
- (D) Assume the metric is wrong because the model is modern.

Q8. Short Answer

Compare an exact method and an approximate or heuristic method relevant to video hierarchy and shot representation. When would the approximate method be justified?

Q9. Calculation

A video has 30 fps and 90,000 frames. How many minutes is it? If one keyframe is selected every 10 seconds, how many keyframes?

Q10. Long Question

Design retrieval for a 2,000-hour sports video archive where users search for goals, fouls and player appearances.

1. Specify shot boundary and scene/event detection.
2. Choose keyframe, motion, audio and transcript features.
3. Build a temporal index and ranking method.
4. Add user feedback for missed events.
5. Evaluate precision@k and event-level recall.



W8 Large Scale Retrieval and BoVW

Pitfall: pixels as words; Focus: local descriptors; **Pitfall: no geometry check;** Focus: visual vocabulary

Q1. MCQ

Which statement is most appropriate for BoVW and semantic annotation?

- (A) BoVW stores raw pixel grids as words.
- (B) The visual vocabulary is made from document titles.
- (C) BoVW eliminates all semantic gap issues.
- (D) BoVW maps local descriptors to visual words and builds sparse histograms.

Q2. Short Answer

Explain BoVW and semantic annotation using the course pipeline language. Your answer must mention keypoints, visual vocabulary, inverted visual index, semantic gap.

Q3. Long Question Semester 1 2024 Final (modified)

Adapt the source-style question to BoVW and semantic annotation. Answer all parts:

1. State the underlying principle.
2. Describe the implementation steps.
3. Give two advantages and two limitations.
4. Propose one concrete improvement for a modern multimedia retrieval setting.

Q4. True/False

True or false: In Large Scale Retrieval and BoVW, higher numerical score always means a better retrieval result. Justify your answer with one counterexample.

Q5. Short Answer

A student makes this mistake: saying BoVW stores pixels rather than local descriptor assignments. Explain why it is wrong and give a corrected approach.

Q6. Table/Diagram

Construct a two-column comparison table for BoVW and semantic annotation: left column = method/component, right column = retrieval benefit and limitation.

Q7. MCQ

A retrieval system for BoVW and semantic annotation performs poorly on a new domain. What is the best first diagnostic step?

- (A) Delete the evaluation set and report only training results.
- (B) Inspect false positives and false negatives by feature, query type and user intent.
- (C) Add more algorithms without checking data or metrics.
- (D) Assume the metric is wrong because the model is modern.

Q8. Short Answer

Compare an exact method and an approximate or heuristic method relevant to BoVW and semantic annotation. When would the approximate method be justified?

Q9. Calculation

An image has visual word histogram (2, 0, 3) and query histogram (1, 1, 2). Compute histogram intersection.

Q10. Long Question

Design BoVW-based landmark image search for mobile queries under scale and viewpoint changes.

1. Define keypoint detection and descriptor extraction.
2. Build the visual vocabulary and image histograms.
3. Choose inverted visual index and geometric verification.
4. Address quantisation error and semantic gap.
5. Evaluate retrieval speed and top-k accuracy.



W9 Recommendation Systems

Pitfall: CF under cold start; **Focus:** hybrid design; **Pitfall:** sparsity; **Focus:** Pearson mean-centering

Q1. MCQ

Which statement is most appropriate for recommendation algorithms?

- (A) Content-based recommendation is safer when items have features but few user ratings.
- (B) User-based CF is best when no users have rated items.
- (C) Cold start only affects search engines, not recommenders.
- (D) Pearson correlation ignores common ratings.

Q2. Short Answer

Explain recommendation algorithms using the course pipeline language. Your answer must mention content-based, collaborative filtering, Pearson, cold start, hybrid.

Q3. Long Question Semester 1 2022 Final (modified)

Adapt the source-style question to recommendation algorithms. Answer all parts:

1. State the underlying principle.
2. Describe the implementation steps.
3. Give two advantages and two limitations.
4. Propose one concrete improvement for a modern multimedia retrieval setting.

Q4. True/False

True or false: In Recommendation Systems, higher numerical score always means a better retrieval result. Justify your answer with one counterexample.

Q5. Short Answer

A student makes this mistake: choosing collaborative filtering for a sparse cold-start system. Explain why it is wrong and give a corrected approach.

Q6. Table/Diagram

Construct a two-column comparison table for recommendation algorithms: left column = method/component, right column = retrieval benefit and limitation.

Q7. MCQ

A retrieval system for recommendation algorithms performs poorly on a new domain. What is the best first diagnostic step?

- (A) Delete the evaluation set and report only training results.
- (B) Add more algorithms without checking data or metrics.
- (C) Inspect false positives and false negatives by feature, query type and user intent.
- (D) Assume the metric is wrong because the model is modern.

Q8. Short Answer

Compare an exact method and an approximate or heuristic method relevant to recommendation algorithms. When would the approximate method be justified?

Q9. Calculation

Alice's mean rating is 4.0. Two neighbours have similarities 0.8 and 0.2; their item deviations are 1 and -2. Predict with weighted CF.

Q10. Long Question

Design recommendations for an online bookstore with 10 million books but only 1,000 active raters.

1. Choose content-based, user-CF, item-CF or hybrid.
2. Define item/user representations and similarity.
3. Handle user and item cold start.
4. Add context or diversity without hurting relevance.
5. Evaluate rating prediction and top-k recommendation.



W10 Information and Video Summarisation

Pitfall: salience without diversity; Focus: coverage; **Pitfall: incoherent skim;** Focus: redundancy control

Q1. MCQ

Which statement is most appropriate for summaries for browsing and retrieval?

- (A) A dynamic snippet is independent of the query.
- (B) A good summary balances coverage and redundancy reduction.
- (C) A video skim is always a still image.
- (D) Summarisation is unrelated to retrieval UX.

Q2. Short Answer

Explain summaries for browsing and retrieval using the course pipeline language. Your answer must mention static/dynamic snippets, keyframes, skims, coverage, redundancy.

Q3. Long Question Semester 1 2024 Final (modified)

Adapt the source-style question to summaries for browsing and retrieval. Answer all parts:

1. State the underlying principle.
2. Describe the implementation steps.
3. Give two advantages and two limitations.
4. Propose one concrete improvement for a modern multimedia retrieval setting.

Q4. True/False

True or false: In Information and Video Summarisation, higher numerical score always means a better retrieval result. Justify your answer with one counterexample.

Q5. Short Answer

A student makes this mistake: optimising salience but ignoring redundancy and coherence. Explain why it is wrong and give a corrected approach.

Q6. Table/Diagram

Construct a two-column comparison table for summaries for browsing and retrieval: left column = method/component, right column = retrieval benefit and limitation.

Q7. MCQ

A retrieval system for summaries for browsing and retrieval performs poorly on a new domain. What is the best first diagnostic step?

- (A) Delete the evaluation set and report only training results.
- (B) Add more algorithms without checking data or metrics.
- (C) Assume the metric is wrong because the model is modern.
- (D) Inspect false positives and false negatives by feature, query type and user intent.

Q8. Short Answer

Compare an exact method and an approximate or heuristic method relevant to summaries for browsing and retrieval. When would the approximate method be justified?

Q9. Calculation

Four candidate shots S_1 – S_4 add new concept counts (5, 2, 0, 4) to the current summary. With equal cost and no redundancy penalty, which shot does greedy max-new-coverage select first?

Q10. Long Question

Design video summarisation for long lecture recordings where students want a 3-minute preview.

1. Define shot/slide/transcript units.
2. Choose importance signals and redundancy control.
3. Decide static keyframes vs dynamic skim.
4. Personalise summaries by query or course topic.
5. Evaluate coverage, coherence and user usefulness.



W11 Social Multimedia and Advanced Topics

Pitfall: popularity bias; **Focus:** content-user-context; **Pitfall:** privacy ignored; **Focus:** fusion normalisation

Q1. MCQ

Which statement is most appropriate for social multimedia retrieval?

- (A) Popularity alone is an unbiased relevance label.
- (B) Personalisation never affects privacy.
- (C) Social retrieval can combine content, user and context signals.
- (D) All social tags are clean ground truth.

Q2. Short Answer

Explain social multimedia retrieval using the course pipeline language. Your answer must mention content, user, context, social graph, multimodal fusion.

Q3. Long Question Semester 1 2024 Final (modified)

Adapt the source-style question to social multimedia retrieval. Answer all parts:

1. State the underlying principle.
2. Describe the implementation steps.
3. Give two advantages and two limitations.
4. Propose one concrete improvement for a modern multimedia retrieval setting.

Q4. True/False

True or false: In Social Multimedia and Advanced Topics, higher numerical score always means a better retrieval result. Justify your answer with one counterexample.

Q5. Short Answer

A student makes this mistake: using popularity as relevance without bias/privacy checks. Explain why it is wrong and give a corrected approach.

Q6. Table/Diagram

Construct a two-column comparison table for social multimedia retrieval: left column = method/component, right column = retrieval benefit and limitation.

Q7. MCQ

A retrieval system for social multimedia retrieval performs poorly on a new domain. What is the best first diagnostic step?

- (A) Inspect false positives and false negatives by feature, query type and user intent.
- (B) Delete the evaluation set and report only training results.
- (C) Add more algorithms without checking data or metrics.
- (D) Assume the metric is wrong because the model is modern.

Q8. Short Answer

Compare an exact method and an approximate or heuristic method relevant to social multimedia retrieval. When would the approximate method be justified?

Q9. Calculation

A social video has relevance score 0.7, popularity score 0.9 and trust score 0.5. Compute late fusion $0.6R + 0.2P + 0.2T$.

Q10. Long Question

Design social multimedia search for a campus event platform containing posts, images, short videos, likes and comments.

1. Define content, user and context signals.
2. Choose multimodal fusion and ranking strategy.
3. Control popularity bias, privacy and spam.
4. Add personalisation with user control.
5. Evaluate relevance, diversity and fairness.



W12 Integration and Exam Design

Pitfall: vague use AI; Focus: measurable innovation; **Pitfall: no ablation;** Focus: hybrid architecture

Q1. MCQ

Which statement is most appropriate for end-to-end retrieval system design?

- (A) A list of algorithms with no scenario is sufficient.
- (B) Evaluation can be skipped for open-ended designs.
- (C) Innovation means only saying 'use AI'.
- (D) A strong design answer links requirements, representation, index, ranking and evaluation.

Q2. Short Answer

Explain end-to-end retrieval system design using the course pipeline language. Your answer must mention requirements, architecture, indexing, ranking, evaluation, innovation.

Q3. Long Question Semester 1 2021 Final (modified)

Adapt the source-style question to end-to-end retrieval system design. Answer all parts:

1. State the underlying principle.
2. Describe the implementation steps.
3. Give two advantages and two limitations.
4. Propose one concrete improvement for a modern multimedia retrieval setting.

Q4. True/False

True or false: In Integration and Exam Design, higher numerical score always means a better retrieval result. Justify your answer with one counterexample.

Q5. Short Answer

A student makes this mistake: listing components without explaining trade-offs. Explain why it is wrong and give a corrected approach.

Q6. Table/Diagram

Construct a two-column comparison table for end-to-end retrieval system design: left column = method/component, right column = retrieval benefit and limitation.

Q7. MCQ

A retrieval system for end-to-end retrieval system design performs poorly on a new domain. What is the best first diagnostic step?

- (A) Delete the evaluation set and report only training results.
- (B) Inspect false positives and false negatives by feature, query type and user intent.
- (C) Add more algorithms without checking data or metrics.
- (D) Assume the metric is wrong because the model is modern.

Q8. Short Answer

Compare an exact method and an approximate or heuristic method relevant to end-to-end retrieval system design. When would the approximate method be justified?

Q9. Calculation

A system improves MAP from 0.32 to 0.40 but latency from 120 ms to 500 ms. State the relative MAP gain and the design trade-off.

Q10. Long Question

Design an end-to-end next-generation retrieval system for USYD Library that supports text, image and video queries.

1. State requirements and query modes.
2. Choose representations for each modality.
3. Combine lexical, visual and semantic indexes.
4. Propose one innovative but measurable improvement.
5. Define metrics, ablations and deployment risks.



Answer Key

W1 Introduction and Retrieval System Design

- Q1: A.
 Q2: Mention the named concepts, then connect representation, indexing, similarity/ranking and evaluation.
 Q3: Cover domain needs, multimedia data types, indexing/ranking choice, evaluation and future improvement; use the source as style only.
 Q4: False. Distances are better when smaller, similarities are better when larger, and some scores require normalisation.
 Q5: The mistake is confusing metadata search with content-based search; corrected answer should use the appropriate representation, normalisation, metric or evaluation context.
 Q6: A good table names concrete components for multimedia retrieval pipeline, gives one benefit and one limitation each, and links them to retrieval effectiveness or efficiency.
 Q7: C.
 Q8: Exact methods maximise correctness but may be slow; approximate methods are justified for large-scale or high-dimensional search when recall/latency is validated.
 Q9: Precision@10 = 8/10 = 0.8. Recall@10 = 8/20 = 0.4.
 Q10: Key points: separate student/staff needs; combine OCR/text, thumbnails/visual features and metadata; inverted index plus vector reranking; feedback from clicks/saves; evaluate nDCG/P@k and latency; note copyright/access constraints.

W2 Text Information Retrieval

- Q1: B.
 Q2: Mention the named concepts, then connect representation, indexing, similarity/ranking and evaluation.
 Q3: Include preprocessing, dictionary/postings construction, multimedia-index limitation and an improvement such as visual words or multimodal metadata.
 Q4: False. Distances are better when smaller, similarities are better when larger, and some scores require normalisation.
 Q5: The mistake is using raw term frequency without IDF or normalisation; corrected answer should use the appropriate representation, normalisation, metric or evaluation context.
 Q6: A good table names concrete components for document preprocessing and inverted index, gives one benefit and one limitation each, and links them to retrieval effectiveness or efficiency.
 Q7: D.
 Q8: Exact methods maximise correctness but may be slow; approximate methods are justified for large-scale or high-dimensional search when recall/latency is validated.
 Q9: $idf = \log_{10} 5 \approx 0.699$. Scores: $2(0.699) = 1.398$ and $6(0.699) = 4.194$.
 Q10: Key points: tokenisation/stemming choices; preserve domain terms; positional postings for phrase/proximity; TF-IDF or BM25 with length normalisation; cautious expansion; evaluate MAP/nDCG and inspect long-document/boilerplate failures.

W3 Web Information Retrieval

- Q1: C.
 Q2: Mention the named concepts, then connect representation, indexing, similarity/ranking and evaluation.
 Q3: Include PageRank/HITS principle, iteration steps, use cases, topic drift/spam issues and a modern freshness/personalisation improvement.
 Q4: False. Distances are better when smaller, similarities are better when larger, and some scores require normalisation.
 Q5: The mistake is forgetting outdegree and teleportation in PageRank; corrected answer should use the appropriate representation, normalisation, metric or evaluation context.
 Q6: A good table names concrete components for PageRank and HITS, gives one benefit and one limitation each, and links them to retrieval effectiveness or efficiency.
 Q7: A.
 Q8: Exact methods maximise correctness but may be slow; approximate methods are justified for large-scale or high-dimensional search when recall/latency is validated.
 Q9: $PR(A) = (1 - 0.85)/3 + 0.85(PR(C)/1) = 0.05 + 0.85/3 \approx 0.333$.
 Q10: Key points: crawler frontier and duplicate control; PageRank with outdegree and teleportation; anchor text as external evidence; HITS for query-topic authority/hub analysis; spam/topic drift mitigation; metrics include relevance and freshness.

W4 Multimedia Basics

- Q1: D.
 Q2: Mention the named concepts, then connect representation, indexing, similarity/ranking and evaluation.
 Q3: Include sampling, quantisation, aliasing, histogram equalisation or acquisition quality, plus retrieval impact.
 Q4: False. Distances are better when smaller, similarities are better when larger, and some scores require normalisation.



Q5: The mistake is describing signal processing without retrieval consequences; corrected answer should use the appropriate representation, normalisation, metric or evaluation context.

Q6: A good table names concrete components for digitisation and media representation, gives one benefit and one limitation each, and links them to retrieval effectiveness or efficiency.

Q7: B.

Q8: Exact methods maximise correctness but may be slow; approximate methods are justified for large-scale or high-dimensional search when recall/latency is validated.

Q9: CDF for 0–3 is (0.1, 0.3, 0.6, 1.0), so mappings are (1, 2, 4, 7).

Q10: Key points: record resolution/bit depth/frame rate/compression; anti-aliasing and normalisation; choose features robust to acquisition variation; metadata for device/quality; compare retrieval before/after preprocessing.

W5 Image Retrieval Features

Q1: A.

Q2: Mention the named concepts, then connect representation, indexing, similarity/ranking and evaluation.

Q3: Include distance formula, feature-vector comparison, ranking by smaller distance and limitation of low-level visual similarity.

Q4: False. Distances are better when smaller, similarities are better when larger, and some scores require normalisation.

Q5: The mistake is treating visually similar low-level features as semantic equality; corrected answer should use the appropriate representation, normalisation, metric or evaluation context.

Q6: Query image is transformed into visual descriptors; index narrows candidates; ranking orders by similarity. Failure may come from semantic gap, illumination change, or wrong feature weighting.

Q7: C.

Q8: Exact methods maximise correctness but may be slow; approximate methods are justified for large-scale or high-dimensional search when recall/latency is validated.

Q9: $\sqrt{0.2^2 + (-0.1)^2 + (-0.2)^2} = \sqrt{0.09} = 0.3$.

Q10: Key points: colour/texture/shape/local descriptors; normalise feature scales; ANN or inverted visual candidates; rerank with better metric/geometric checks; relevance feedback/deep semantics; evaluate P@k under lighting/viewpoint changes.

W6 CBIR Indexing and Feedback

Q1: B.

Q2: Mention the named concepts, then connect representation, indexing, similarity/ranking and evaluation.

Q3: Include exact-vs-ANN indexing trade-off, k-d/LSH limits in high dimension, feature normalisation and validation with recall@k/latency.

Q4: False. Distances are better when smaller, similarities are better when larger, and some scores require normalisation.

Q5: The mistake is mixing incomparable feature scales; corrected answer should use the appropriate representation, normalisation, metric or evaluation context.

Q6: A good table names concrete components for similarity search and indexing, gives one benefit and one limitation each, and links them to retrieval effectiveness or efficiency.

Q7: D.

Q8: Exact methods maximise correctness but may be slow; approximate methods are justified for large-scale or high-dimensional search when recall/latency is validated.

Q9: X has $0.5(0.8) + 0.5(20/100) = 0.50$. Y has $0.5(0.4) + 0.5(70/100) = 0.55$. Smaller is better, so X ranks first.

Q10: Key points: exact search baseline on subset; ANN/LSH/vector graph for candidates; min-max or z-score feature fusion; rerank top K; feedback reweights features; report recall@k, P95 latency, memory and update cost.

W7 Video Retrieval

Q1: C.

Q2: Mention the named concepts, then connect representation, indexing, similarity/ranking and evaluation.

Q3: Include video hierarchy, shot-boundary methods, keyframe vs shot representation, streaming limitations and hybrid improvement.

Q4: False. Distances are better when smaller, similarities are better when larger, and some scores require normalisation.

Q5: The mistake is using keyframes when temporal motion is essential; corrected answer should use the appropriate representation, normalisation, metric or evaluation context.

Q6: Frame level indexes local/global visual features; shot level indexes keyframes and motion; scene/event level indexes semantic labels, transcript, audio and temporal context.

Q7: A.

Q8: Exact methods maximise correctness but may be slow; approximate methods are justified for large-scale or high-dimensional search when recall/latency is validated.

Q9: Duration $90000/30 = 3000$ seconds = 50 minutes. Keyframes $3000/10 = 300$.



Q10: Key points: shot boundary using histogram/edge/motion cues; keyframes plus motion/audio/transcript; feedback on missed goals/fouls; evaluate event-level recall, P@k and boundary errors.

W8 Large Scale Retrieval and BoVW

Q1: D.

Q2: Mention the named concepts, then connect representation, indexing, similarity/ranking and evaluation.

Q3: Include keypoints, visual vocabulary, sparse histograms, inverted visual index, quantisation/geometric verification and semantic gap.

Q4: False. Distances are better when smaller, similarities are better when larger, and some scores require normalisation.

Q5: The mistake is saying BoVW stores pixels rather than local descriptor assignments; corrected answer should use the appropriate representation, normalisation, metric or evaluation context.

Q6: A good table names concrete components for BoVW and semantic annotation, gives one benefit and one limitation each, and links them to retrieval effectiveness or efficiency.

Q7: B.

Q8: Exact methods maximise correctness but may be slow; approximate methods are justified for large-scale or high-dimensional search when recall/latency is validated.

Q9: $\min(2, 1) + \min(0, 1) + \min(3, 2) = 1 + 0 + 2 = 3$.

Q10: Key points: repeatable keypoints; descriptors clustered into visual vocabulary; sparse histograms in inverted index; geometric verification; address codebook/quantisation errors; evaluate top-k accuracy and mobile latency.

W9 Recommendation Systems

Q1: A.

Q2: Mention the named concepts, then connect representation, indexing, similarity/ranking and evaluation.

Q3: Include CF/content-based principles, cold-start/sparsity, similarity choice and hybrid recommendation justification.

Q4: False. Distances are better when smaller, similarities are better when larger, and some scores require normalisation.

Q5: The mistake is choosing collaborative filtering for a sparse cold-start system; corrected answer should use the appropriate representation, normalisation, metric or evaluation context.

Q6: A good table names concrete components for recommendation algorithms, gives one benefit and one limitation each, and links them to retrieval effectiveness or efficiency.

Q7: C.

Q8: Exact methods maximise correctness but may be slow; approximate methods are justified for large-scale or high-dimensional search when recall/latency is validated.

Q9: $4 + (0.8(1) + 0.2(-2)) / (|0.8| + |0.2|) = 4 + 0.4 = 4.4$.

Q10: Key points: sparse ratings favour content-based or hybrid; item metadata and user profiles; Pearson/cosine only when enough overlap; cold-start onboarding/popularity priors; evaluate RMSE plus top-k precision/diversity.

W10 Information and Video Summarisation

Q1: B.

Q2: Mention the named concepts, then connect representation, indexing, similarity/ranking and evaluation.

Q3: Include keyframe/skim summary, coverage-vs-redundancy, content/social signals, and evaluation by usefulness/coherence.

Q4: False. Distances are better when smaller, similarities are better when larger, and some scores require normalisation.

Q5: The mistake is optimising salience but ignoring redundancy and coherence; corrected answer should use the appropriate representation, normalisation, metric or evaluation context.

Q6: A good table names concrete components for summaries for browsing and retrieval, gives one benefit and one limitation each, and links them to retrieval effectiveness or efficiency.

Q7: D.

Q8: Exact methods maximise correctness but may be slow; approximate methods are justified for large-scale or high-dimensional search when recall/latency is validated.

Q9: Select S_1 , because it adds 5 new concepts, the largest marginal coverage among (5, 2, 0, 4).

Q10: Key points: segment by slides/shots/transcript; score importance from topic, novelty and user query; reduce redundancy; choose keyframe summary vs skim; evaluate coverage, coherence and student usefulness.

W11 Social Multimedia and Advanced Topics

Q1: C.

Q2: Mention the named concepts, then connect representation, indexing, similarity/ranking and evaluation.

Q3: Include content-user-context signals, social bias/privacy risks, multimodal fusion and personalised/social summary design.

Q4: False. Distances are better when smaller, similarities are better when larger, and some scores require normalisation.



Q5: The mistake is using popularity as relevance without bias/privacy checks; corrected answer should use the representation, normalisation, metric or evaluation context.

Q6: A good table names concrete components for social multimedia retrieval, gives one benefit and one limitation each, links them to retrieval effectiveness or efficiency.

Q7: A.

Q8: Exact methods maximise correctness but may be slow; approximate methods are justified for large-scale or high-dimensional search when recall/latency is validated.

Q9: $0.6(0.7) + 0.2(0.9) + 0.2(0.5) = 0.42 + 0.18 + 0.10 = 0.70$.

Q10: Key points: combine media features, tags/comments, user graph and context; late/early fusion with normalisation; privacy and bias controls; personalisation transparency; evaluate relevance, diversity, fairness and spam robustness.

W12 Integration and Exam Design

Q1: D.

Q2: Mention the named concepts, then connect representation, indexing, similarity/ranking and evaluation.

Q3: Include requirements, heterogeneous media architecture, index/ranking/fusion, evaluation and measurable innovation.

Q4: False. Distances are better when smaller, similarities are better when larger, and some scores require normalisation.

Q5: The mistake is listing components without explaining trade-offs; corrected answer should use the appropriate representation, normalisation, metric or evaluation context.

Q6: A good table names concrete components for end-to-end retrieval system design, gives one benefit and one limitation each, and links them to retrieval effectiveness or efficiency.

Q7: B.

Q8: Exact methods maximise correctness but may be slow; approximate methods are justified for large-scale or high-dimensional search when recall/latency is validated.

Q9: MAP gain $(0.40 - 0.32)/0.32 = 25\%$. It may be unacceptable for interactive search unless reranking is limited/cached.

Q10: Key points: requirements first; text OCR/transcripts, image/video embeddings and metadata; hybrid lexical/vector architecture; innovation tied to measurable limitation; ablation by modality; deployment risks include privacy, copyright and latency.